

Individual Assignment 5

Kelsey Hammond

2026-05-10

1. Set up

Packages

```
# general use and cleaning
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

# wrapping axis labels
library(scales)

# reading in .xlsx files
library(readxl)

# visualizing missing data
library(naniar)

# arranging plots
library(patchwork)

# data visualization
library(NatParksPalettes)
```

Data

```
# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))

# aquatic invertebrates
aq_ins <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Aquatic Insects")

# site information
sites <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "sites")

# water quality information
water_quality <- read_xlsx(here("data",
  "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Water Quality",
  na = c("", "n/a", "N/A", "over", "173+"))
```

Cleaning Data

New sites object

```
# creating new object from sites
ncos_sites <- sites |>
  # filter to only include NCOS sites sampled four times a year
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")
```

Cleaning taxon_list

```
# creating new clean object from taxon_list
taxon_list_clean <- taxon_list |>
  # converting taxon name to snake case (no spaces or capital letters,
  # only underscores)
  mutate(verbatim_name = to_any_case(verbatim_name, case = "snake"))
```

Cleaning water_quality

```
# creating new clean object from water quality
water_quality_clean <- water_quality |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites in ncos_sites
  semi_join(ncos_sites, by = "site") |>
  # create new column to join with later data frames
  unite("date_site", date, site, remove = TRUE) |>
  # group by date_site column
  group_by(date_site) |>
  # calculate median pH, salinity, DO
  summarize(
    med_ph = median(p_h, na.rm = TRUE),
    med_sal = median(salinity_ppt, na.rm = TRUE),
    med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
  # ungroup data frame
  ungroup() |>
  # filter out salinity outliers
  filter(date_site != "2022-08-12_NVBR")
```

Cleaning aq_ins

```
# creating new clean object from aquatic inverts
aq_ins_clean <- aq_ins |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites occurring in ncos_sites
  semi_join(ncos_sites, by = c("site")) |>
  # renaming columns
  rename(date = date_on_vial) |>
  # select columns of interest
  select(
    date, sample_type, site,
    ostracod,
    copepod,
    hemiptera_corixidae_boatman,
    cladocera,
    nematode,
    diptera_ceratopogonidae,
    annelida_oligochaete,
```

```

    diptera_chironomid,
    annelida_polychaete) |>
# filter to only include "filtered beaker" samples
filter(sample_type %in% c("FB 250", "FB250")) |>
# convert the data frame to long format
pivot_longer(cols = ostracod:annelida_polychaete,
              names_to = "taxon",
              values_to = "abundance") |>
# group by date, site, taxon
group_by(date, site, taxon) |>
# calculate average abundance per liter
# (sum abundance divided by 7.5 liters)
summarize(ave_lit = sum(abundance, na.rm = TRUE)/7.5) |>
# ungroup data frame
ungroup() |>
# create a new column called `date_site` to join with water quality
unite("date_site", date, site, remove = FALSE) |>
# join with water quality
left_join(water_quality_clean, by = c("date_site")) |>
# join with taxon list
left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
# add full names of sites
mutate(site_full = case_when(
  site == "NVBR" ~ "North of Venoco Bridge",
  site == "NEC" ~ "South of Venoco Bridge",
  site == "NMC" ~ "Main Channel",
  site == "NPB" ~ "South of Phelps Bridge",
  site == "NPB1" ~ "North of Phelps Bridge",
  site == "NPB2" ~ "Phelps Road",
  site == "NWP" ~ "West Pond",
  site == "NDC" ~ "Devereux Creek"
)) |>
# setting factor levels for site
mutate(site_full = fct_reorder(site_full,
                               med_sal,
                               .fun = "median", .na_rm = TRUE),
       site_full = fct_rev(site_full))

```

Class code

```
# base layer
salinity <- ggplot(data = aq_ins_clean,
  # x-axis: site
  # y-axis: mean salinity
  mapping = aes(x = site_full,
    y = med_sal)) +
# first layer: salinity measurements at each survey
geom_jitter(height = 0,
  width = 0.1,
  shape = 21) +
# wrapping x-axis text
scale_x_discrete(labels = label_wrap(width = 10)) +
# second layer: points to represent means
stat_summary(fun = median,
  color = "red",
  size = 1) +
# relabelling x- and y-axes
labs(x = "Site",
  y = "Median salinity (ppt)",
  title = "Site salinity") +
# different theme
theme_bw()
```

```
# creating new object from clean aquatic inverts
copepods <- aq_ins_clean |>
# filter to only include copepods
filter(taxon == "copepod") |>
# log + 1 transform mean abundance per liter
mutate(log_ave_lit = log(ave_lit + 1))
```

```
# base layer
copepods <- ggplot(data = copepods,
  # x-axis: site
  # y-axis: mean abundance/L
  mapping = aes(x = site_full,
    y = log_ave_lit)) +
# first layer: points to represent observations
geom_jitter(height = 0,
  shape = 21,
```

```

        width = 0.1) +
# second layer: points to represent medians
stat_summary(geom = "point",
             fun = "median",
             size = 4,
             color = "blue") +
# wrapping x-axis text
scale_x_discrete(labels = label_wrap(10)) +
# relabelling x- and y-axis, adding title
labs(x = "Site",
     y = "log + 1 transformed average abundance/L",
     title = "Copepods") +
# different theme
theme_bw()

```

2. Problems

1. Visualizing differences across sites in major taxon abundance

a. Plan your figure

- create an object to store new plot
- x-axis should be `site_full`
- y-axis should be `log_ave_lit`
- geometry to represent average abundance/L: `geom_jitter()`
- geometry to represent medians: `stat_summary()`

b. Wrangle your data

```

# creating new object from clean aquatic inverts
ostracods <- aq_ins_clean |>
# filtering to only include ostracods
filter(taxon == "ostracod") |>
# log + 1 transform mean abundance per liter
mutate(log_ave_lit = log(ave_lit + 1))

# displaying a sample of 10 observations
slice_sample(ostracods, n = 10)

```

A tibble: 10 x 24

date_site	date	site	taxon	ave_lit	med_ph	med_sal	med_do
-----------	------	------	-------	---------	--------	---------	--------

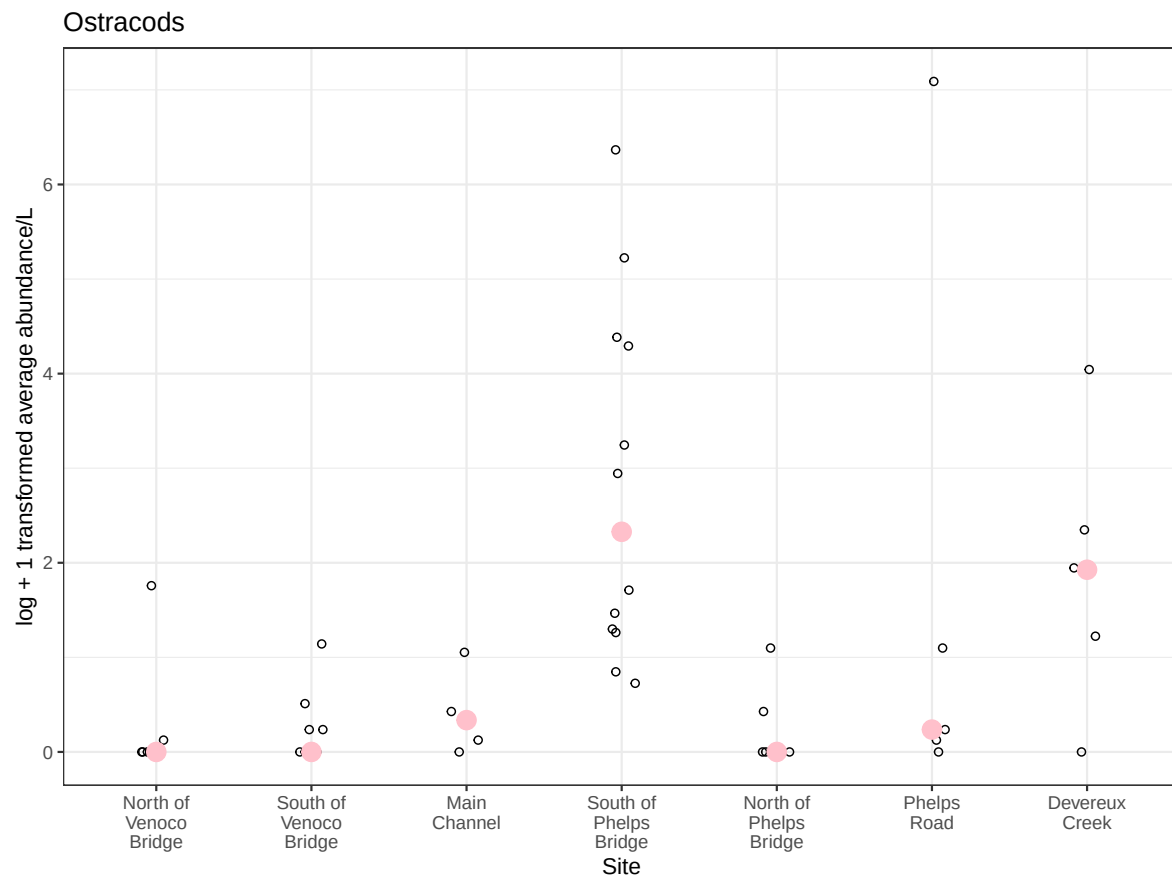
	<chr>	<dtm>	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1	2021-11-16_NPB	2021-11-16 00:00:00	NPB	ostr~	72.1	7.3	2.38	12.1
2	2023-01-28_NPB1	2023-01-28 00:00:00	NPB1	ostr~	0	7.66	NA	9.74
3	2020-02-21_NVBR	2020-02-21 00:00:00	NVBR	ostr~	4.8	7.34	4.1	6.1
4	2022-01-21_NEC	2022-01-21 00:00:00	NEC	ostr~	0	NA	NA	NA
5	2023-05-05_NPB	2023-05-05 00:00:00	NPB	ostr~	24.7	NA	NA	12.1
6	2023-02-28_NPB	2023-02-28 00:00:00	NPB	ostr~	2.53	NA	NA	NA
7	2023-08-12_NEC	2023-08-12 00:00:00	NEC	ostr~	0	NA	NA	NA
8	2022-02-21_NPB	2022-02-21 00:00:00	NPB	ostr~	1.33	7.39	34.1	0.34
9	2022-11-18_NVBR	2022-11-18 00:00:00	NVBR	ostr~	0	NA	NA	NA
10	2023-11-14_NPB	2023-11-14 00:00:00	NPB	ostr~	18	NA	NA	NA

i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
comments <chr>, site_full <fct>, log_ave_lit <dbl>

c. Code your figure

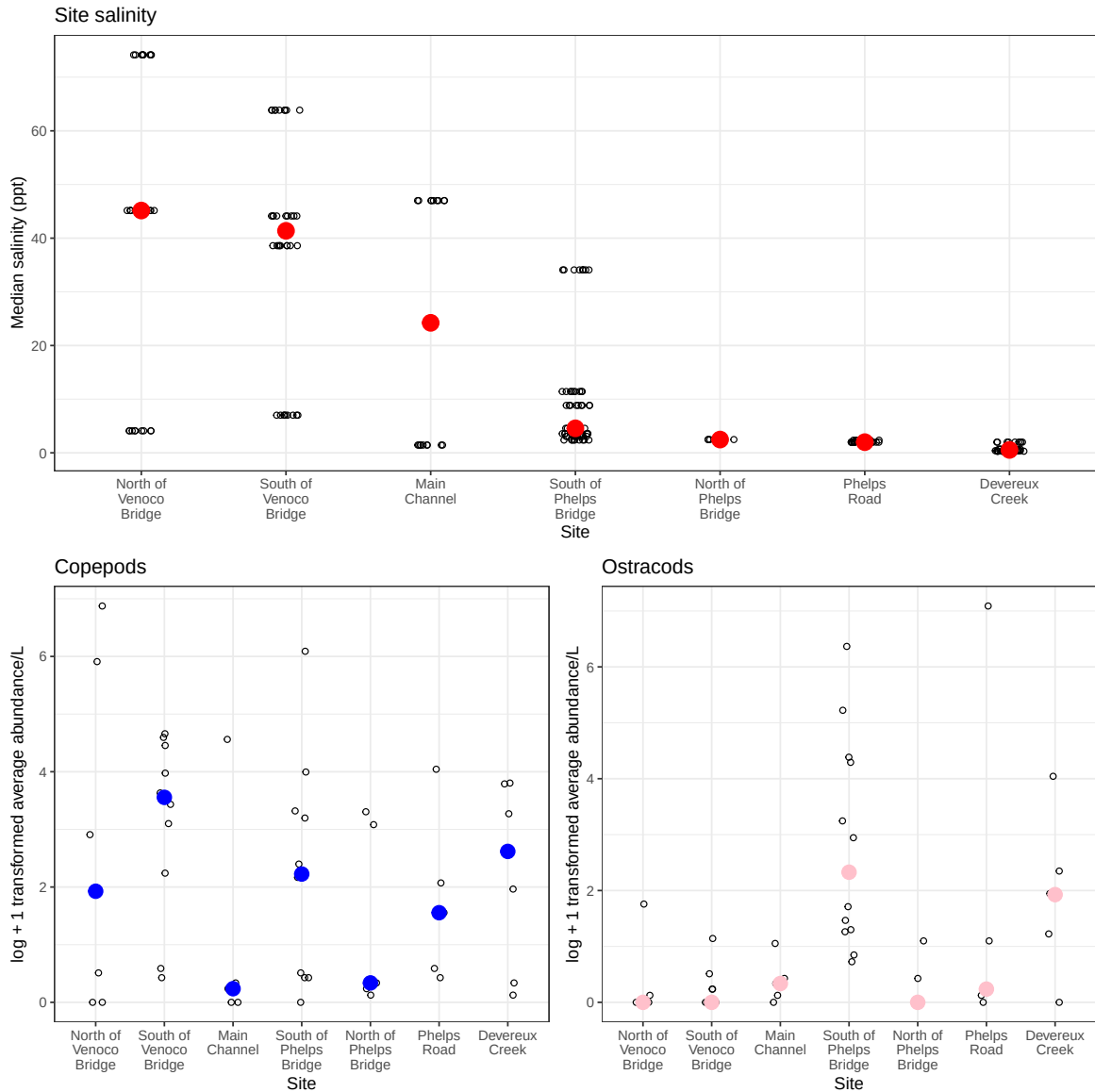
```
# creating new object to store ostracod plot
ostracod_plot <- ggplot(data = ostracods,
  # x-axis: full site name
  # y-axis: log transformed average abundance/L
  mapping = aes(x = site_full,
    y = log_ave_lit)) +
  # first layer: points to represent observations
  geom_jitter(height = 0,
    width = 0.1,
    shape = 21) +
  # second layer: points to represent median
  stat_summary(geom = "point",
    fun = "median",
    size = 4,
    color = "pink") +
  # wrapping x-axis labels to fit
  scale_x_discrete(labels = label_wrap(10)) +
  # relabeling axes and adding title
  labs(x = "Site",
    y = "log + 1 transformed average abundance/L",
    title = "Ostracods") +
  # custom theme
  theme_bw()
```

```
# displaying plot
ostracod_plot
```



d. Create a multipanel figure

```
# combining salinity, copepod, and ostracod plots into one figure
salinity / (copepods | ostracod_plot)
```

e. Interpret your figure

Copepod abundance is high at North of Venoco Bridge, South of Venoco Bridge, South of Phelps Bridge, and Devereux Creek. It is low at Main Channel, North of Phelps Bridge, and Phelps Road.

Ostracod abundance is high at South of Phelps Bridge and Devereux Creek. It is low at North of Venoco Bridge, South of Vencoco Bridge, Main Channel, North of Phelps Bridge, and Phelps Road.

The two highest levels of ostracod abundance are at South of Phelps Bridge and Devereux Creek. When looking at the median salinity for those two sites, they are both very low, Devereux Creek being the lowest. This may show that ostracods prefer low salinity.

2. Visualizing total abundance as a function of salinity

a. Plan your figure

- x-axis: med_sal
- y-axis: log_ave_lit
- color: taxon
- geometry: geom_point()
- facets: facet_wrap(~taxon, scales = "free")

b. Wrangle your data

```
# editing the data frame
aq_ins_clean <- aq_ins_clean |>
  # changing taxon to be title case
  mutate(taxon = to_any_case(taxon, case = "title")) |>
  # log + 1 transforming average abundance/L
  mutate(log_ave_lit = log(ave_lit + 1)) |>
  # selecting only average abundance, taxon, and median salinity columns
  select(log_ave_lit, taxon, med_sal) |>
  # reordering taxon by maximum abundance
  mutate(taxon = fct_reorder(taxon, log_ave_lit, .fun = "max", .na_rm = TRUE))

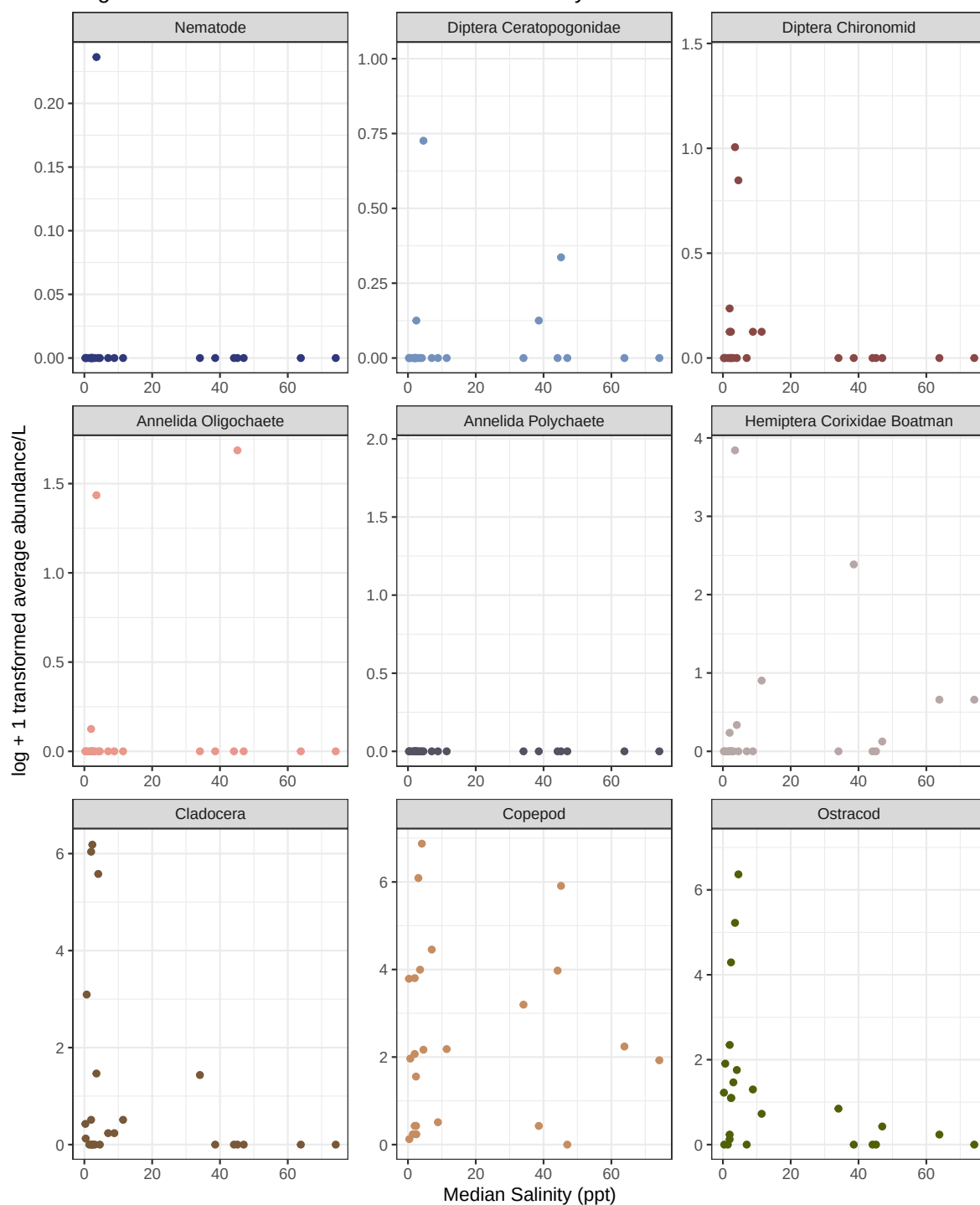
# displaying sample of 10 observations
slice_sample(aq_ins_clean, n = 10)
```

```
# A tibble: 10 x 3
  log_ave_lit taxon          med_sal
    <dbl> <fct>          <dbl>
1     0.903 Diptera Ceratopogonidae NA
2     0     Nematode NA
3     0     Ostracod 38.6
4     0     Annelida Polychaete 2.00
5     1.44 Diptera Chironomid NA
6     0     Nematode NA
7     0     Nematode 1.45
8     1.93 Annelida Polychaete NA
9     4.66 Copepod NA
10    1.93 Copepod 74.2
```

c. Code your figure

```
# base layer: ggplot
ggplot(data = aq_ins_clean,
       # x-axis: median salinity
       # y-axis: log average abundance/L
       # color: taxon
       mapping = aes(x = med_sal,
                     y = log_ave_lit,
                     color = taxon)) +
# first layer: point
geom_point() +
# relabelling axes, adding title
labs(x = "Median Salinity (ppt)",
     y = "log + 1 transformed average abundance/L",
     title = "Log transformed abundance as a function of salinity") +
# custom theme
theme_bw() +
# custom colors
scale_color_manual(values = natparks.pals("Torres")) +
# removing legend
theme(legend.position = "none") +
# wrapping so axes are on free scales
facet_wrap(~taxon, scales = "free")
```

Log transformed abundance as a function of salinity



d. Interpret your figure

Nematodes, Annelida Oligochaete, and Annelida Polychaete all have similar relationships between salinity and abundance with the abundance remaining relatively low despite changes in salinity.

Cladocera, Copepod, and Ostracod all have similar relationships to each other with high abundance when there is low salinity. Copepod abundance varies slightly from the other two with high abundance around median salinity of 40 ppt. Himiptera Corixidae Boatman also follows a similar trend but with a weaker relationship.

Diptera Ceratopogonidae, Diptera Chironomid have similar relationship, both have a slight increase at median salinity of 40 ppt (similar to Copepod).